

PSD-Rayleigh Directional Memory Tails

A Relative Gramian Bound for Fourth-Mode Support

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Abstract

The right-frame reduction of RH-113 leaves one missing quantity: the tail acting on four recent right singular directions. We prove a positive-tail cross-Gram inequality and turn it into a multiplicative exterior-volume bound. If $T \succeq 0$, $\|T\| \leq \delta$, P is a packet isometry, Q is a right four-frame, and $R = (I - PP^*)TPQ$, then

$$R^*R \preceq \delta Q^*P^*TPQ.$$

More generally, any PSD upper $D \succeq R^*R$ yields a generalized Rayleigh constant γ and a four-volume factor $(1 - \gamma)_+^4$. The five-scale audit has zero failures; on the fine chain the packet-block bound reduces the worst scalar gamma from 0.00134724 to 0.000175802. This is a directional theorem and finite audit, not an all-level physical law.

1 The directional problem

Let A be a recent finite-memory cross action and $K = A + R$ its full action on a packet. RH-113 showed that a four-column action $Y = AQ$ gives a lower bound for the full fourth exterior norm. A scalar error $\|R\| \leq \delta$ leads to a weak-mode penalty because the relative constant is bounded by $\delta/s_4(Y)$. The purpose of this paper is to preserve the Gramian of the tail in the same four-dimensional frame.

We work in a Hilbert space, with P an isometry from packet coordinates into the ambient domain and $Q : \mathbb{C}^4 \rightarrow \text{ran } P$ an isometry. Write $\Pi = PP^*$ and let $T \succeq 0$ be a discarded memory tail. Its directional cross action is

$$R = (I - \Pi)TPQ, \quad Y = (I - \Pi)APQ.$$

All matrix inequalities below are in the usual Loewner order; the exterior volume identities use the standard singular-value calculus [Horn and Johnson, 1991].

2 Positive-tail cross-Gram theorem

Theorem 1 (Positive-tail cross-Gram theorem). *If $T \succeq 0$ and $\|T\| \leq \delta$, then*

$$R^*R \preceq \delta Q^*P^*TPQ. \tag{1}$$

Proof. Since $0 \preceq I - \Pi \preceq I$,

$$T(I - \Pi)T \preceq T^2 \preceq \delta T.$$

Compress this inequality on the left and right by PQ to obtain (1). No commutation between T and the packet projection is needed. $\square$

The scalar upper $R^*R \preceq \delta^2 I$ follows from $\|R\| \leq \delta$. Equation (1) is stronger when the tail has little energy in the current packet directions. It is also compatible with matrix-free action: only the four columns TPQ and the 4×4 packet block are required.

3 Relative Rayleigh volume theorem

Let $G = Y^*Y$ be positive definite and suppose $D \succeq R^*R$. Define

$$\gamma^2 = \lambda_{\max}(G^{-1/2}DG^{-1/2}).$$

Theorem 2 (Relative Rayleigh theorem). *If $\gamma < 1$, then*

$$(Y + R)^*(Y + R) \succeq (1 - \gamma)^2 G, \quad D_4(Y + R) \geq (1 - \gamma)^4 D_4(Y). \quad (2)$$

For arbitrary γ , the second lower bound is understood with $(1 - \gamma)_+^4$.

Proof. The definition of γ is equivalent to $R^*R \preceq \gamma^2 G$. Therefore $\|Rx\| \leq \gamma \|Yx\|$ for every x . The reverse triangle inequality gives $\|(Y + R)x\| \geq (1 - \gamma) \|Yx\|$ when $\gamma < 1$, which is the first Loewner inequality. Determinants are monotone on positive definite 4×4 matrices, so taking square roots of determinants gives the second claim. $\square$

Corollary 3 (Positive-tail certificate). *Under Theorem 1, use $D = \delta Q^*P^*TPQ$ in Theorem 2. If $U \geq \|K\|$, then the normalized fourth exterior volume obeys*

$$\frac{\|\bigwedge^4 K\|_2}{U^4} \geq (1 - \gamma)_+^4 \frac{D_4(Y)}{U^4}.$$

4 Sharpness and comparison

The factor is sharp: take $R = -\gamma Y$, so $R^*R = \gamma^2 G$ and $D_4(Y + R) = (1 - \gamma)^4 D_4(Y)$. The scalar specialization has

$$\gamma_{\text{scalar}} \leq \frac{\delta}{s_4(Y)}, \quad D_{\text{scalar}} = \delta^2 I.$$

The packet-block version replaces the isotropic identity by the tail energy seen by PQ . It can be better by orders of magnitude, but it cannot be used unless that PSD block is itself controlled or computed with a validated upper enclosure.

5 Five-scale audit

We recompute the RH-110 finite-memory chain at five scales, two channels, three thresholds, and 360 records. At each update we form the positive tail Gramian, the recent top-four frame, and three upper choices: scalar $\delta^2 I$, positive packet-block $\delta Q^* P^* T P Q$, and the exact directional residual Gramian. The analytic tail radius is enlarged only by outward floating-point guards; PSD corrections are recorded separately.

There are zero block-dominance and certificate failures. On the 234 fine records, the largest generalized constants are

upper choice	minimum γ	maximum γ	fine support at 10^{-8}
scalar $\delta^2 I$	1.2571×10^{-12}	0.00134724	78
positive packet block	3.83×10^{-159}	0.000175802	78
exact directional Gram	0	6.59362×10^{-6}	78

Table 1: Fine-chain generalized tail constants. The exact lower endpoint is an audit reference; the packet-block line is the structured theorem route.

The three certificates have the same fine threshold counts: 78, 72, and 55 for thresholds 10^{-8} , 10^{-6} , and 10^{-4} , respectively. This is not a failure of the theorem: the archived thresholds are already crossed by the product-Weyl lower bound. The gain is quantitative slack and a route to future scales where the scalar bound may cease to be usable.

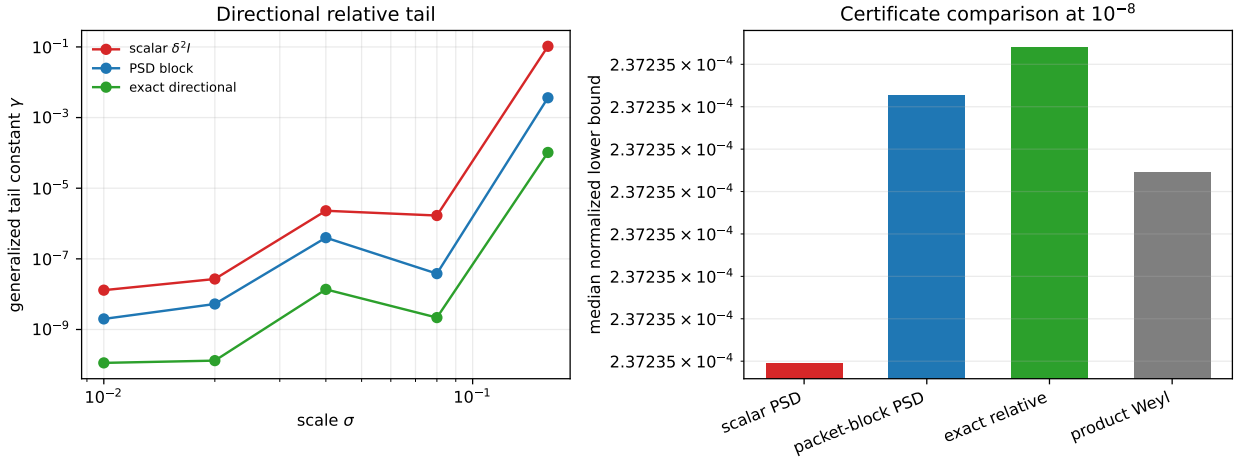


Figure 1: Left: median generalized tail constants by scale. Right: median normalized lower certificates on the finest threshold.

6 Route consequence and boundary

RH-114 turns a vague directional-tail hope into a precise sufficient condition: control one 4×4 PSD block and one generalized eigenvalue. The positive-tail proof is unconditional for finite

PSD tails. What remains open is an all-level physical bound on that block, compatible with memory depth and the changing right frame. RH-115 will compose the scalar, packet-block, trace-concentration, and capacity certificates and keep the strongest valid lower endpoint.

No all-level exterior law, uniform Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis conclusion is claimed.

References

Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.